

EA51: Empirical Analyses

Spring 2026

4

Course Description

Empirical Analyses focuses on thinking creatively. You will become familiar with the central methods used in the natural and social sciences, enabling you to frame problems effectively, develop and test hypotheses, and engage in informed conjecture.

An overview of the Foundation Year curriculum, including the schedule of Units, Lessons, and Big Questions for this course, can be found on the [Cornerstone Curriculum Map](#) on the Hub.

Note: this syllabus is subject to change.

Course Objectives & Learning Outcomes

Biases

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

#biasmitigation: Identify methods to mitigate the effect of biases and determine when it is appropriate to do so. (H) EA

Research Design

#casestudy: Design and interpret case studies. (C) EA

#comparisongroups: Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA

#interventionalstudy: Design and interpret experimental studies. (C) EA

#interviewsurvey: Design and interpret primary research performed as interviews or surveys (individually or in groups). (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#studyreplication: Evaluate and incorporate replicability in empirical study designs. (C) EA

Science of Science Communication

#audience: Tailor oral and written work by considering the situation and perspective of the people receiving it. (H) MC

#conformity: Identify and analyze social norms, their features, origins and development; evaluate the appropriateness of conformity and contrarianism and apply strategies of social change when applicable. (H) CX

#evidencebased: Identify and appropriately structure the information needed to support an argument effectively. (H) MC

Uphold Course Policies and Guidelines

#professionalism: Follow established guidelines to present yourself and your work products professionally. (H) MC

#responsibility: Follow through on commitments; be proactive and adaptable. (H) CX

#ClassParticipation: Contribute to a productive learning environment by attending class prepared and on time, participating actively in class sessions, and applying learning outcomes through in-class assessments.

Prerequisites & Working Knowledge

There are no additional prerequisites beyond satisfactory completion of Cornerstone summer preparatory work which can be found on [The Hub](#).

Assignments

Cornerstone assignments are original works (e.g., essays, short papers, problem sets, or models) that consolidate the student's work on a combined set of HCs. All Minerva courses include at least one location-based assignment (LBA) that involves engaging in an activity in the student's current city of residence and provides an opportunity to apply HCs in a new, practical, real-world, context. All assignments are graded against a mandatory set of targeted HCs that must be addressed. Students may also address other HCs introduced earlier in this, or any other, Cornerstone course. The expectations regarding which HCs are targeted and how they will be scored are outlined in the assignment instructions for each assignment. The way in which assignment scores contribute to the final course grades can be found in the Assessment section of the Cornerstone syllabi.

Note: Sunday is considered the beginning of the academic week for determining due dates.

Assignment Title	Weight	Due	Released	Closed
Biases	7x	Thu, Week 4	Tue of Week 2	Sun of Week 5
Research Design	7x	Sat, Week 9	Wed of Week 6	Tue of Week 10
Science of Science Communication	5x	Sat, Week 13	Sat of Week 11	Sun of Week 13
Science of Science Communication - INDIVIDUAL component	2x	Fri, Week 14	Fri of Week 11	Sat of Week 14

Course Scoring

Type	% of Course Score
Assignments	60%
Biases	20%
Research Design	20%
Science of Science Communication - INDIVIDUAL component	5%
Science of Science Communication	15%
Classroom Scores	40%
Pre-Class Work	13%
Poll Responses & Makeup Work	12%
Verbal Contributions	15%
Breakouts	0%

Required Texts

Boots, E., Draffin-Jacquín, F., Chia, Z. (2022, Aug). Approaching Dense Readings. Minerva Academic Team.
<https://docs.google.com/document/d/1gvxmiXtM4jXWtxNbRZaTs8gaGA2-49F4gAfFbrqw9MQ/edit?usp=sharing>

Minerva Academic Team. (2023). HC Resources.
<https://my.minerva.edu/academics/hc-resources/>

Schedule of Topics & Readings

This course meets for 2 class sessions each week.

Unit 1: Biases

Unaided, unsystematic human observation is not a good way of constructing reliable knowledge. In this unit, we review the biases to which we are subject, such as top-down and bottom-up factors that influence our perception and attention, emotional influences, preconceived ideas, and biases resulting from use of heuristics. The goal is to identify how these can influence decision making and the encoding and recall of memories. We use the Big Question “How reliable is testimony?” to provide context throughout this unit.

Session 1: (1.1) Heuristics and biases: When do they work?

Learning Outcomes

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

Reading, Videos, and Other Preparation Resources

(Review). Read the description and example for **#heuristics** in the HC Handbook

<https://my.minerva.edu/academics/hc-resources/hc-handbook/heuristics/>

Read the description and example for **#biasidentification** in the HC Handbook:

<https://my.minerva.edu/academics/hc-resources/hc-handbook/biasidentification/>

Behavioral Science Solutions (n.d.). Read the following brief sections:

Overview

1. Heuristics: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/heuristic/>
2. Fast and Frugal: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/fast-and-frugal/>

Focus on these new heuristics

1. Representativeness heuristic: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/representativeness-heuristic/>
2. Availability heuristic: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/availability-heuristic/>
3. Anchoring heuristic: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/anchoring-heuristic/>
4. Affect heuristic: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/affect-heuristic/>
5. Recognition heuristic: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/recognition-heuristic/>
6. Take-the-best heuristic: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/take-the-best-heuristic/>
7. Satisficing heuristic: <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/satisficing/>

For each of these heuristics, focus on what the rule for simplifying decision making is.

<https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/heuristic/>

Brush, J. E. (2015). Decision-making shortcuts: The good and the bad. *New England Journal of Medicine Knowledge Plus*.

<https://blogs.jwatch.org/cardioexchange/2011/09/19/decision-making-shortcuts-the-good-and-the-bad/>

Behavioral Science Solutions (n.d.). Cognitive bias. In *Behavioral Science Concepts*.

<https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/cognitive-bias/>

Cincinnati Law. (2018, November 20). *Implicit Bias, Professor Song L. Richardson* [Video]. YouTube.

<https://www.youtube.com/watch?v=SzFQDAsqZw>

Dror, I. E. (2018). Biases in forensic experts. *Science*, 360(6386), 243–243.

<https://science.sciencemag.org/content/sci/360/6386/243.full.pdf>

(Review). Dosmann, A., & Stein, G. (2019). *Common Learning Techniques: Using #scienceoflearning to Make the Most of Your Study Time*. Minerva University.

https://docs.google.com/document/d/1gDJ8DMbIgKNzb_xK71i3WsnPMF7hbsfBBnn4LPiBnLk/edit?tab=t.0

(Review). Dosmann, A. (2020). *A brief guide to different categories of heuristics*. Minerva University.

https://docs.google.com/document/d/1FsL2FVFfa1t_UD8KeQNqRTQ39EwuadjGM5kgSL-zK0nw/edit?tab=t.0

Session 2: (1.2) Types of Bias 1: Emotional, Attentional, and Memory

Learning Outcomes

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

Reading, Videos, and Other Preparation Resources

Yue, C. (n.d.). *Divided attention, selective attention, inattention blindness, and change blindness* [Video]. Khan Academy.

🔗 <https://www.khanacademy.org/science/health-and-medicine/executive-systems-of-the-brain/attention-language-lesson/v/divided-selective-attention-inattention-change-blindness>

Bodenhausen, G. V., and Hugenberg, K. (2009). Attention, perception, and social cognition. In F. Strack & J. Förster (Eds.), *Social cognition: The basis of human interaction* (pp. 1-22). Psychology Press.

🔗 <https://books.google.com/books?hl=en&lr=&id=qj8tTp39ThMC&oi=fnd&pg=PA1&ots=eLwscxYAjK&sig=0RlkRwo75W7M67NnhShrD7oew5g#v=onepage&q&f=false>

Lerner, J. S., Li, Y., Valdesolo, P., & Kassam, K. S. (2015). Emotion and Decision Making. *Annual Review of Psychology*, 66(1), 799–823.

🔗 <http://www.annualreviews.org/eprint/vVKIPZU5r9dTCzbgdD3M/full/10.1146/annurev-psych-010213-115043>

Damasio, A. (2009, August 11). *When emotions make better decisions* [Video file]. YouTube.

🔗 https://www.youtube.com/watch?v=1wup_K2WN0I

Innocence Project. (2018, November 19). *Human Factors in Wrongful Convictions: Memory Malleability* [Video]. YouTube.

🔗 <https://www.youtube.com/watch?v=jCl66OWjXMM>

Stein, G. (2021). *Memory Bias*. Minerva University.

🔗 https://course-resources.minerva.edu/uploaded_files/mke/00209023-6099/memory-bias-2.jpg

Foong, E. (2014, April 26). *The Misinformation Effect* [Video]. YouTube.

🔗 <https://www.youtube.com/watch?v=iMPIWkFtd88>

Optional: Cherry, K. (2022). How the attentional bias influences the decisions we make. *Verywellmind*.

🔗 <https://www.verywellmind.com/what-is-an-attentional-bias-2795027>

Optional: Looser, C. E. (2019, June). *Forum Magic*. Minerva University.

🔗 https://course-resources.minerva.edu/uploaded_files/mke/00229503-5474/forum-magic.pdf

Session 3: (2.1) Types of Bias 2: Confirmation bias and how heuristics and biases interact

Learning Outcomes

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

Reading, Videos, and Other Preparation Resources

Behavioral Science Solutions. (n.d.). Confirmation bias. In *Behavioral Science Concepts*.

🔗 <https://www.behavioraleconomics.com/resources/mini-encyclopedia-of-be/confirmation-bias/>

Bennett, B. (2014, March 29). *Confirmation bias* [Video]. YouTube.

🔗 https://www.youtube.com/watch?v=B_YkdMwEO5U

Von Bergen, C. W. and Bressler, M. S. (2018). Confirmation Bias in Entrepreneurship. *Journal of Management Policy and Practice*, 19, 74-84. One semester fair use.

🔗 <https://articlegateway.com/index.php/JMPP/article/view/49>

Innocence Project. (n.d.). The Causes.

<http://www.innocenceproject.org/causes-wrongful-conviction>

Read the following sections and cases:

1. *Eyewitness misidentification* <https://www.innocenceproject.org/causes/eyewitness-misidentification/>
2. *The case page for Marvin Anderson* <http://www.innocenceproject.org/cases-false-imprisonment/marvin-anderson>
3. *Misapplication of forensic science* <https://www.innocenceproject.org/causes/misapplication-forensic-science/>

Hammond, C. (2011, March 10). *Fingerprints on trial* [Audio File]. BBC Sounds.

PLEASE NOTE: The recording needs to be downloaded to listen to the original program: <https://www.bbc.co.uk/programmes/b00z5zyc>

A Transcript related to the Shirley Mckie case can be found here: http://news.bbc.co.uk/2/hi/uk_news/scotland/605129.stm

https://course-resources.minerva.edu/uploaded_files/mu/00398183-5523/insidehealth-20110310-fingerprintsontrial.mp3

Kassin, S. M., Dror, I. E., & Kukucka, J. (2013). The forensic confirmation bias: Problems, perspectives, and proposed solutions. *Journal of Applied Research in Memory and Cognition*, 2(1), 42–52. Available in Elsevier Science Direct.

<https://doi.org/10.1016/j.jarmac.2013.01.001>

Review: Foong, E. (2014, April 26). *The Misinformation Effect* [Video]. YouTube.

<https://www.youtube.com/watch?v=iMPIWkFtd88>

Review: Stein, G. (2021). *Memory Bias*. Minerva University.

https://course-resources.minerva.edu/uploaded_files/mke/00209023-6099/memory-bias-2.jpg

Session 4: (2.2) How to mitigate bias 1: Things individuals should do

Learning Outcomes

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

#biasmitigation: Identify methods to mitigate the effect of biases and determine when it is appropriate to do so. (H) EA

Reading, Videos, and Other Preparation Resources

Read the description and example for **#biasmitigation** in the HC Handbook:

<https://my.minerva.edu/academics/hc-resources/hc-handbook/biasmitigation/>

Santos, L. (n.d.). *Cognitive Biases: The GI Joe Fallacy* [Video]. Khan Academy.

<https://www.khanacademy.org/partner-content/wi-phi/wiphi-critical-thinking/wiphi-cognitive-biases/v/gi-joe-fallacy>

Lerner, J. S., Li, Y., Valdesolo, P., & Kassam, K. S. (2015b). Emotion and Decision Making. *Annual Review of Psychology*, 66(1), 799–823.

<http://www.annualreviews.org/eprint/vVKIPZU5r9dTCzbgdD3M/full/10.1146/annurev-psych-010213-115043>

Stafford, T. (2017). How to get people to overcome their bias. *BBC Future*.

<http://www.bbc.com/future/story/20170131-why-wont-some-people-listen-to-reason>

Rasmussen, L. (n.d.). What if pigs really could fly? Get creative to predict future events. *Global Cognition*.

<https://www.globalcognition.org/think-unlikely-possibilities/>

Sieck, W. (n.d.). A scientific reasoning strategy to make everyday discoveries. *Global Cognition*.

<https://www.globalcognition.org/scientific-reasoning-and-discovery/>

Croskerry, P.; Singhal, G.; Mamede, S.; *Cognitive debiasing 2: impediments to and strategies for change*. *BMJ Qual Saf.* 2013 Aug 30;22(Suppl 2):ii65–ii72. doi: 10.1136/bmjqs-2012-001713

<https://pmc.ncbi.nlm.nih.gov/articles/PMC3786644/>

Read the Assignment “Biases” on your dashboard and come prepared to answer and ask questions in class.

Session 5: (3.1) How to mitigate bias 2: Things organizations and institutions should do

Learning Outcomes

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

#biasmitigation: Identify methods to mitigate the effect of biases and determine when it is appropriate to do so. (H) EA

Reading, Videos, and Other Preparation Resources

Cuffari, B. (2019). Forensic confirmation bias. *News Medical Life Sciences*.

🔗 [https://www.news-medical.net/life-sciences/Forensic-Confirmation-Bias-\(Cognitive-Bias-in-Forensics\).aspx](https://www.news-medical.net/life-sciences/Forensic-Confirmation-Bias-(Cognitive-Bias-in-Forensics).aspx)

Kassin, S. M., Dror, I. E., & Kukucka, J. (2013). The forensic confirmation bias: Problems, perspectives, and proposed solutions. *Journal of Applied Research in Memory and Cognition*, 2(1), 42–52. Available in Elsevier Science Direct

Innocence Project. (n.d.). *Eyewitness Identification Reform*.

🔗 <https://innocenceproject.org/in-focus-eyewitness-identification-reforms/>

Schmidt, B. (n.d.). *Gendered Language in Teacher Reviews*. Benschmidt.org.

🔗 <https://benschmidt.org/profGender/#%7B%22database%22%3A%22RMP%22%2C%22plotType%22%3A%22pointchart%22%2C%22method%22>

Flaherty, C. (2019, May 20). *Teaching Evals: Bias and Tenure*. Inside Higher Ed.

🔗 <https://www.insidehighered.com/news/2019/05/20/fighting-gender-bias-student-evaluations-teaching-and-tenures-effect-instruction>

Johnson, S. K. and Kirk, J. F. (2020, March 5). Research: To Reduce Gender Bias, Anonymize Job Applications. *Harvard Business Review*.

🔗 <https://web.archive.org/web/20230325065919/https://hbr.org/2020/03/research-to-reduce-gender-bias-anonymize-job-applications>

Session 6: (3.2) Heuristics and Biases: Review and Transfer

Learning Outcomes

#modeling: Recognize how models can be used to describe a system, explain a set of data, and generate predictions. (C) EA

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

#biasmitigation: Identify methods to mitigate the effect of biases and determine when it is appropriate to do so. (H) EA

Reading, Videos, and Other Preparation Resources

Review previous readings from the unit.

Johnson, K. (2022, March 7). *The Hidden Role of Facial Recognition Tech in Many Arrests*. Wired.

🔗 <https://www.wired.com/story/hidden-role-facial-recognition-tech-arrests/>

Angwin, J., Larson, J., Kirchner, L., Mattu, S., Weychert, R., and Slight, D. (2016, May 23). *Machine Bias*. ProPublica.

🔗 <https://www.propublica.org/article/machine-bias-risk-assessments-in-criminal-sentencing>

Buolamwini, J. (2016, November). *How I'm fighting bias in algorithms*. TED.

🔗 https://www.ted.com/talks/joy_buolamwini_how_i_m_fighting_bias_in_algorithms?language=en

Jeng, K. (2024, October). *Wrongful Convictions Revealed the Dangers of Trusting Unvalidated Science. Are We About to Repeat the Same Mistakes With AI?* Innocence Project.

🔗 <https://innocenceproject.org/news/wrongful-convictions-revealed-the-dangers-of-trusting-unvalidated-science-are-we-about-to-repeat-the-same-mistakes-with-ai/>

Unit 2: Research Design

This unit is a survey of major types of study design applied in the natural and social sciences. We explore data collection for cases in which direct manipulation of the phenomena being studied is not possible (e.g., observational studies, case studies, surveys), and examples in which variables are manipulated to different degrees (e.g., randomized controlled medical trials, quasi-experiments). Emphasis is on the tenets of good research design, strengths and limitations of different design types, generalizability of inferences drawn from different study designs, and practice designing, applying and evaluating research studies.

Session 7: (4.1) Interventional and Observational Studies

Learning Outcomes

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#correlation: Apply and interpret measures of correlation; distinguish correlation and causation. (H) FA

Reading, Videos, and Other Preparation Resources

Lacey, M. (1997). *Experimental design*. Yale University.

<http://www.stat.yale.edu/Courses/1997-98/101/expdes.htm>

University of California Museum of Paleontology. (2016). Understanding Science: Fair tests.

1. A do-it-yourself guide: <https://undsci.berkeley.edu/fair-tests-a-do-it-yourself-guide/>
2. Fair tests in the field of medicine: <https://undsci.berkeley.edu/fair-tests-in-the-field-of-medicine-aiding-alzheimer-patients/>
3. Fair tests in the fossil record: <https://undsci.berkeley.edu/fair-tests-in-the-fossil-record-avoiding-extinction/>
4. Fair tests in physics: <https://undsci.berkeley.edu/fair-tests-in-physics-examining-eclipses/>

McDonald, J.H. (2014). Confounding Variables. In *Handbook of biological statistics* (3rd ed.). Baltimore, MD: Sparky House Publishing.

<http://www.biostathandbook.com/confounding.html>

University of California Museum of Paleontology. (2008). Understanding Evolution.

1. *Introduction to evolution* http://evolution.berkeley.edu/evolibrary/article/0_0_0/evo_02
2. *Resisting our drugs* http://evolution.berkeley.edu/evolibrary/article/bergstrom_03
3. *Battle strategies* <https://evolution.berkeley.edu/battling-bacterial-evolution/battle-strategies/>
4. *Testing the strategies* https://evolution.berkeley.edu/evolibrary/article/0_0_0/bergstrom_05
5. *Turbo-charged evolution* <https://evolution.berkeley.edu/turbo-charged-evolution/>

(Optional). University of California Museum of Paleontology. (2008). Mechanisms: the process of evolution. In *Understanding Evolution*.

http://evolution.berkeley.edu/evolibrary/article/evo_14

Session 8: (4.2) Sampling in observational and interventional studies

Learning Outcomes

#interventionalstudy: Design and interpret experimental studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#hypothesisdevelopment: Evaluate the link between hypothesis-driven research and the theories or observations that motivate it. (C) EA

#constraints: Identify and apply constraint satisfaction as a way to solve problems. (C) EA

#significance: Apply, interpret, and distinguish practical and statistical significance. (C) FA

Reading, Videos, and Other Preparation Resources

Read the description and example for **#sampling** in the Outcome Index:

<https://forum.minerva.edu/app/outcome-index/general-learning-outcomes/sampling>

Bradley, V. C., Kuriwaki, S., Isakov, M., Sejdinovic, D., Meng, X.-L., and Flaxman, S. (2021). Unrepresentative big surveys significantly overestimated US vaccine uptake. *Nature*, 600(7890), 695-699.

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC8653636/pdf/41586_2021_Article_4198.pdf

Dr Nic's Maths and Stats. (2012, March 14). *Sampling: Simple random, convenience, systematic, cluster, stratified* [Video]. YouTube.

<https://www.youtube.com/watch?v=be9e-Q-jC-0>

Penn State Eberly College of Science. (n.d.). Simple random sampling and other sampling methods. In *STAT 100: Statistical Concepts and Reasoning*.

<https://web.archive.org/web/20240926210943/https://online.stat.psu.edu/stat100/lesson/2/2.4>

Azar, B. (2010). Are your findings 'WEIRD'? *Monitor on Psychology*, 41(5). American Psychological Association.

<https://www.apa.org/monitor/2010/05/weird>

Chino, F. (2019, August 21). *Minority Participation in Clinical Trials: A Call to Action*. Milken Institute.

https://milkeninstitute.org/sites/default/files/2022-03/Diversity%20in%20Clinical%20Trials_v6.pdf

[Review] Figueroa, A., de Molière, L., Pegels, A., Never, B., & Kutzner, F. (2019). Show me (more than) the money! Assessing the social and psychological dimensions to energy efficient lighting in Kenya. *Energy Research & Social Science*, 47, 224–232.

<https://doi.org/10.1016/j.erss.2018.10.002>

[OPTIONAL] Reinhart, A. (2015). Pseudoreplication: choose your data wisely. In *Statistics Done Wrong: The woefully complete guide*. No Starch Press.

<https://www.statisticsonewrong.com/pseudoreplication.html>

Session 9: (5.1) Comparison Groups and Controlling for Variables

Learning Outcomes

#comparisongroups: Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#studyreplication: Evaluate and incorporate replicability in empirical study designs. (C) EA

Reading, Videos, and Other Preparation Resources

Read the description and example for **#comparisongroups** in the Outcome Index:

<https://forum.minerva.edu/app/outcome-index/general-learning-outcomes/comparisongroups>

Boots, E., Draffin-Jacquín, F., and Chia, Z. Z. (2022). *Approaching Dense Readings*. Minerva University.

<https://docs.google.com/document/d/1gvxmiXtM4jXWtxNbRZaTs8gaGA2-49F4gAfFbrqw9MQ/edit#heading=h.a0q6s59iiphn>

Wagler, R. (2011). The Impact of Human Activities on Biological Evolution: A Topic of Consideration for Evolution Educators. *Evo Edu Outreach*, 4, 343-347, 2011.

<https://evolution-outreach.biomedcentral.com/articles/10.1007/s12052-011-0326-5>

Carpi, A., and Egger, A.E. (2009). Controlling variables. *Visionlearning POS-2* (3), 2009.

<https://www.visionlearning.com/en/library/Process-of-Science/49/Controlling-Variables/163>

Carpi, A., and Egger, A.E. (2008). Experimentation in scientific research. *Visionlearning POS-1* (7), 2008.

<https://www.visionlearning.com/en/library/Process-of-Science/49/Experimentation-in-Scientific-Research/150>

(Review). Lacey, M. (1997). Experimental design. *Statistics 101-103: Summaries of topics*. Yale University.

<http://www.stat.yale.edu/Courses/1997-98/101/expdes.htm>

(Review). McDonald, J.H. (2014). *Handbook of biological statistics (3rd ed.)*. Baltimore, MD: Sparky House Publishing.

<http://www.biostathandbook.com/confounding.html>

(Review) University of California Museum of Paleontology. (2016). Understanding Science: Fair tests.

<https://undsci.berkeley.edu/fair-tests-in-the-fossil-record-avoiding-extinction/>

Read the description and example for **#studyreplication** in the Outcome Index:

<https://forum.minerva.edu/app/outcome-index/general-learning-outcomes/studyreplication>

Session 10: (5.2) Randomized Controlled Trials

Learning Outcomes

#comparisongroups: Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA

#interventionalstudy: Design and interpret experimental studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#testability: Evaluate whether hypotheses lead to testable predictions. (H) EA

Reading, Videos, and Other Preparation Resources

Sibbald, B., & Roland, M. (1998). Understanding controlled trials: Why are randomised controlled trials important? *British Medical Journal*, 316(7126), 201.

<http://www.ncbi.nlm.nih.gov/pmc/articles/PMC2665449/pdf/9468688.pdf>

Levin, K. A. (2007). Study design VII. Randomised controlled trials. *Evidence-Based Dentistry*, 8(1), 22-23.

<https://www.nature.com/articles/6400473>

Newman, T. (2017). Placebos: The power of the placebo effect. *MedicalNewsToday*.

<https://www.medicalnewstoday.com/articles/306437.php>

Natural history group. (2017, February 22). In *Wikipedia*.

https://en.wikipedia.org/wiki/Natural_history_group

Kaptchuk, T. J., Stason, W. B., Davis, R. B., Legedza, A. R., Schnyer, R. N., Kerr, C. E., ... & Goldman, R. H. (2006). Sham device v inert pill: randomised controlled trial of two placebo treatments. *BMJ*, 332(7538), 391-397.

<http://www.bmj.com/content/bmj/332/7538/391.full.pdf>

(Review). Chino, F. (2019, August 21). *Minority Participation in Clinical Trials: A Call to Action*. Milken Institute.

https://milkeninstitute.org/sites/default/files/2022-03/Diversity%20in%20Clinical%20Trials_v6.pdf

Read through the assignment: Research Design on your dashboard and be prepared to ask and answer questions about it in class.

Session 11: (6.2) Observational study design

Learning Outcomes

#comparisongroups: Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#studyreplication: Evaluate and incorporate replicability in empirical study designs. (C) EA

Reading, Videos, and Other Preparation Resources

Sagarin, R., and Pauchard, A. (2010). Observational approaches in ecology open new ground in a changing world. *Frontiers in Ecology and the Environment*, 8(7), 379-386.

https://course-resources.minerva.edu/uploaded_files/mu/00247169-0705/saaharin.pdf

University of California Museum of Paleontology. (2016). Acidic oceans prompt evolution. In *Understanding Evolution*.

http://evolution.berkeley.edu/evolibrary/news/121001_acidicoceans

Besen, J., and Gan, S. D. (2014). A critical evaluation of clinical research study designs. *Journal of Investigative Dermatology*, 134(3), e18.

[http://www.jidonline.org/article/S0022-202X\(15\)36682-3/pdf](http://www.jidonline.org/article/S0022-202X(15)36682-3/pdf)

Session 12: (7.1) Quasi-experimental design

Learning Outcomes

#comparisongroups: Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#studyreplication: Evaluate and incorporate replicability in empirical study designs. (C) EA

Reading, Videos, and Other Preparation Resources

Harris, A. D., Bradham, D. D., Baumgarten, M., Zuckerman, I. H., Fink, J. C., & Perencevich, E. N. (2004). The use and interpretation of quasi-experimental studies in infectious diseases. *Clinical Infectious Diseases*, 38(11), 1586-1591.

https://www.researchgate.net/publication/8550796_The_Use_and_Interpretation_of_Quasi-Experimental_Studies_in_Infectious_Diseases

McFarland, C., Pow, M., Glick, J. (2018, September 19). Quasi Experimentation at Netflix. *The Netflix Tech Blog*.

<https://netflixtechblog.com/quasi-experimentation-at-netflix-566b57d2e362>

Netflix. (2020, September 7). *My Octopus Teacher | Official Trailer* [Video]. YouTube.

<https://www.youtube.com/watch?v=3s0LTDhqe5A>

Read the Research Design Assignment on your Dashboard.

Session 13: (7.2) Study Replication

Learning Outcomes

#interventionalstudy: Design and interpret experimental studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#studyreplication: Evaluate and incorporate replicability in empirical study designs. (C) EA

#hypothesisdevelopment: Evaluate the link between hypothesis-driven research and the theories or observations that motivate it. (C) EA

Reading, Videos, and Other Preparation Resources

Nakagawa, S., & Parker, T. H. (2015). Replicating research in ecology and evolution: feasibility, incentives, and the cost-benefit conundrum. *BMC biology*, 13(1), 88.

<https://bmcbiol.biomedcentral.com/articles/10.1186/s12915-015-0196-3>

Law, W. & Salick, J. (2005) Human-induced dwarfing of Himalayan snow lotus, *Saussurea laniceps* (Asteraceae). *PNAS*

<https://www.pnas.org/doi/full/10.1073/pnas.0502931102>

Singal, J. (2018, December 18). Psychology research is still fixated on a tiny fraction of humans - here's how to fix that. *The British Psychology Society*.

<https://www.bps.org.uk/research-digest/psychology-research-still-fixated-tiny-fraction-humans-heres-how-fix>

(Review) Read the description and example for **#studyreplication** in the HC Handbook:

<https://my.minerva.edu/academics/hc-resources/hc-handbook/studyreplication/>

Session 14: (8.1) Synthesis: Interventional and Observational Studies

Learning Outcomes

#comparisongroups: Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#hypothesisdevelopment: Evaluate the link between hypothesis-driven research and the theories or observations that motivate it. (C) EA

#constraints: Identify and apply constraint satisfaction as a way to solve problems. (C) EA

#significance: Apply, interpret, and distinguish practical and statistical significance. (C) FA

#testability: Evaluate whether hypotheses lead to testable predictions. (H) EA

Reading, Videos, and Other Preparation Resources

University of California Museum of Paleontology. (2008). Sexual selection. In *Understanding Evolution*.

http://evolution.berkeley.edu/evolibrary/article/evo_28

Allendorf, F. W., & Hard, J. J. (2009). Human-induced evolution caused by unnatural selection through harvest of wild animals. *Proceedings of the National Academy of Sciences*, 106 (Supplement 1), 9987-9994.

http://www.pnas.org/content/106/Supplement_1/9987.full

Review: Besen, J., & Gan, S. D. (2014). A critical evaluation of clinical research study designs. *Journal of Investigative Dermatology*, 134(3), e18.

[http://www.jidonline.org/article/S0022-202X\(15\)36682-3/pdf](http://www.jidonline.org/article/S0022-202X(15)36682-3/pdf)

Session 15: (8.2) Case study

Learning Outcomes

#casestudy: Design and interpret case studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

#biasmitigation: Identify methods to mitigate the effect of biases and determine when it is appropriate to do so. (H) EA

#variables: Identify and classify the relevant variables of a system, problem, or model. (H) FA

Reading, Videos, and Other Preparation Resources

Donnelly, Katie. (2022, February 8). *Defining Case Studies as a Research Method*. Minerva University.

https://docs.google.com/document/d/1_z7-4x65UBKOrlomyFcFfE1yNsk8nH8cOVpR0B2A9zE/edit?tab=t.0

Hamilton, J. (2017, May 21). Why Brain Scientists Are Still Obsessed With The Curious Case Of Phineas Gage. *NPR Weekend Edition Sunday*.

<https://www.npr.org/sections/health-shots/2017/05/21/528966102/why-brain-scientists-are-still-obsessed-with-the-curious-case-of-phineas-gage>

American Scientific Frontiers (2010). *Memory Loss - A Case Study (Patient EP)* [Video]. YouTube.

https://www.youtube.com/watch?v=CL-0gfpf_6g

Calvet, G., Aguiar, R., Melo, A., Sampaio, S., Filippis, I.D., Fabri, A., ... Filippis, A.M. (2016). Detection and sequencing of Zika virus from amniotic fluid of fetuses with microcephaly in Brazil: a case study. *Lancet Infectious Disease*, 16: 653–660. (1 semester fair use)

https://course-resources.minerva.edu/uploaded_files/mu/00255606-9396/calvetetal-2016-zikavirus-casestudy-annotated--1-.pdf

Crowe, S., Cresswell, K., Robertson, A., Huby, G., Avery, A., & Sheikh, A. (2011). The case study approach. *BMC Medical Research Methodology*, 11(1), 1.

<http://bmcmmedresmethodol.biomedcentral.com/articles/10.1186/1471-2288-11-100>

PBS. (1999). *HIV Immunity*. PBS Learning Media.

<https://pbsnc.pbslearningmedia.org/resource/tdc02.sci.life.gen.hivimmunity/hiv-immunity/>

Read the description and example for **#casestudy** in the HC Handbook:

<https://my.minerva.edu/academics/hc-resources/hc-handbook/casestudy/>

Session 16: (9.1) Synthesis: Research Design

Learning Outcomes

#casestudy: Design and interpret case studies. (C) EA

#comparisongroups: Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#sampling: Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#heuristics: Identify when to use heuristics and when to avoid them. (H) EA

Reading, Videos, and Other Preparation Resources

(Review). Lacey, M. (1997). Experimental design. *Statistics 101-103: Summaries of topics*. Yale University.

<http://www.stat.yale.edu/Courses/1997-98/101/expdes.htm>

(Review). Besen, J., & Gan, S. D. (2014). A critical evaluation of clinical research study designs. *Journal of Investigative Dermatology*, 134(3), e18.

[http://www.jidonline.org/article/S0022-202X\(15\)36682-3/pdf](http://www.jidonline.org/article/S0022-202X(15)36682-3/pdf)

McDowell, I. (n.d.). Study designs. *Society, the individual, and medicine*.

https://web.archive.org/web/20180219225520/http://www.med.uottawa.ca/SIM/data/Study_Designs_e.htm

(Review). Crowe, S., Cresswell, K., Robertson, A., Huby, G., Avery, A., & Sheikh, A. (2011). The case study approach. *BMC Medical Research Methodology*, 11(1), 1.

<http://bmcmmedresmethodol.biomedcentral.com/articles/10.1186/1471-2288-11-100>

(Review). Statistics Learning Centre. (2012). *Sampling: Simple random, convenience, systematic, cluster, stratified* [Video]. YouTube.

<https://www.youtube.com/watch?v=be9e-Q-jC-0>

(Review) Donnelly, Katie. (2022, February 8). *Defining Case Studies as a Research Method*. Minerva University.

https://docs.google.com/document/d/1_z7-4x65UBKOrlomyFcFfE1yNsk8nH8cOVpR0B2A9zE/edit?tab=t.0

(Optional, but recommended) Nolen, S. (2022, August 16). *Forgotten Virus: Zika Families and Researchers Struggle for Support*. *The New York Times*.

<https://www.nytimes.com/2022/08/16/health/zika-children-research.html>

Session 17: (9.2) Interviews and Surveys 1

Learning Outcomes

- #interventionalstudy:** Design and interpret experimental studies. (C) EA
- #interviewsurvey:** Design and interpret primary research performed as interviews or surveys (individually or in groups). (C) EA
- #observationalstudy:** Design and interpret observational studies. (C) EA
- #correlation:** Apply and interpret measures of correlation; distinguish correlation and causation. (H) FA

Reading, Videos, and Other Preparation Resources

_Qualitative vs. quantitative research: What's the difference? _(n.d.). Survey Monkey.

<https://www.surveymonkey.com/mp/quantitative-vs-qualitative-research/>

Thayer-Hart, N., Dykema, J., Elver, K., Schaeffer, N. C., & Stevenson, J. (2010). *Survey fundamentals: A guide to designing and implementing surveys*. Office of Quality Improvement.

<https://www.lonestar.edu/departments/honorsprogram/A%20Guide%20to%20Designing%20and%20Implementing%20Surveys.pdf>

Driscoll, D. L., Appiah-Yeboah, A., Salib, P., and Rupert, D. J. (2007). Merging qualitative and quantitative data in mixed methods research: How to and why not. *Ecological and Environmental Anthropology*. University of Georgia, Paper 18.

<http://digitalcommons.unl.edu/icwdmeeaa18>

Read the description and example for **#interviewsurvey** in the HC Handbook:

<https://my.minerva.edu/academics/hc-resources/hc-handbook/interviewsurvey/>

Session 18: (10.1) Interviews and Surveys 2

Learning Outcomes

- #interviewsurvey:** Design and interpret primary research performed as interviews or surveys (individually or in groups). (C) EA
- #observationalstudy:** Design and interpret observational studies. (C) EA

Reading, Videos, and Other Preparation Resources

Burns, K. E. A., & Kho, M. E. (2015, April 7). *How to assess a survey report: a guide for readers and peer reviewers*. Canadian Medical Association Journal.

<https://www.cmaj.ca/content/187/6/e198>

Faverio, M., Anderson, M., & Park, E. (2025, April 22). *Teens, Social Media and Mental Health*. Pew Research Center.

<https://www.pewresearch.org/internet/2025/04/22/teens-social-media-and-mental-health/>

Session 19: (10.2) Research Design to Public Recommendations

Learning Outcomes

- #casestudy:** Design and interpret case studies. (C) EA
- #comparisongroups:** Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA
- #interventionalstudy:** Design and interpret experimental studies. (C) EA
- #observationalstudy:** Design and interpret observational studies. (C) EA
- #sampling:** Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA
- #studyreplication:** Evaluate and incorporate replicability in empirical study designs. (C) EA
- #correlation:** Apply and interpret measures of correlation; distinguish correlation and causation. (H) FA

#testability: Evaluate whether hypotheses lead to testable predictions. (H) EA

Reading, Videos, and Other Preparation Resources

Pan, A., Sun, Q., Bernstein, A. M., Schulze, M. B., Manson, J. E., Stampfer, M. J., Hu, F. B. (2012). Red meat consumption and mortality: results from 2 prospective cohort studies. *Archives of Internal Medicine*, 172(7), 555-563

<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC3712342/>

Lamotte, S. (2019, September). Red and processed meat are OK to eat, controversial new guidelines claim. Don't believe it, leading experts say. *CNN Health*.

<https://edition.cnn.com/2019/09/30/health/red-meat-low-quality-evidence-controversy-wellness/index.html>

Messerli, F. (2012). Chocolate consumption, cognitive function, and Nobel laureates. *The New England Journal of Medicine*, 367, 1562-1564.

<https://scottbarrykaufman.com/wp-content/uploads/2012/10/Messerli-2012.pdf>

(Optional). Aubrey, A. (2012). Death By Bacon? Study Finds Eating Meat Is Risky. *National Public Radio*.

<http://www.npr.org/sections/thesalt/2012/03/12/148457233/death-by-bacon-study-finds-eating-meat-is-risky>

Unit 3: Science of Science Communication

An important application of problem solving skills, bias recognition and mitigation, and research design is the communication of scientific findings to a variety of audiences. In this unit, you will analyze specific audiences and defend what scientific information needs to be presented (and what does not need to be presented) to particular audiences to accomplish your goals.

Session 20: (12.1) Introduction to Science of Science Communication

Learning Outcomes

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#plausibility: Evaluate whether hypotheses are based on plausible premises or assumptions. (H) EA

#testability: Evaluate whether hypotheses lead to testable predictions. (H) EA

#evidencebased: Identify and appropriately structure the information needed to support an argument effectively. (H) MC

Reading, Videos, and Other Preparation Resources

Oreskes, N. (2014, June 25). *Why we should trust scientists* [Video]. TED Talks.

https://www.ted.com/talks/naomi_oreskes_why_we_should_trust_scientists?language=en

Shermer, M. (2009). *Baloney detection kit* [Video]. YouTube.

<https://www.youtube.com/watch?v=aNSHZG9bIQQ&t=83s>

Tretkoff, E. (2007). This month in physics history, September 1904: Robert Wood debunks N-rays. *APS News*.

<https://www.aps.org/publications/apsnews/200708/history.cfm>

Brunning, A. (2015). A rough guide to spotting bad science. *Compound Chem*.

<https://www.compoundchem.com/2014/04/02/a-rough-guide-to-spotting-bad-science/>

Climate Action Plan 2021 | San Francisco Environment Department (SFE). (n.d.).

<https://www.sfenvironment.org/media/14441>

Read the Assignment "Science of Science Communication" on your dashboard.

TBWA/Media Arts Lab, & Apple. (2023, September 21). *2030 Status Mother Nature Apple*. [Video]. AdAge.

<https://adage.com/video/2030-status-mother-nature-apple/>

Session 21: (12.2) Audience Analysis

Learning Outcomes

#audience: Tailor oral and written work by considering the situation and perspective of the people receiving it. (H) MC

#gapanalysis: Identify and evaluate whether there are suitable existing solutions to a problem or whether a creative new solution is required. (C) EA

#evidencebased: Identify and appropriately structure the information needed to support an argument effectively. (H) MC

Reading, Videos, and Other Preparation Resources

Huertas, A. (2016, June 9). *There is no general public: Starting with audience for stronger science communication plans*. Medium.

<https://medium.com/science-communication-media/there-is-no-general-public-starting-with-audience-for-stronger-science-communication-plans-6603a81da8d6>

Global Warming's Six Audiences around the world - Yale Program on Climate Change Communication. (2024, August 7). Yale Program on Climate Change Communication.

<https://climatecommunication.yale.edu/publications/six-audiences-around-the-world/>

New Guidebook: Using the Six Americas Super Short Survey (SASSY) in Campaigns and Education - Yale Program on Climate Change Communication. (2025, August 7). Yale Program on Climate Change Communication.

<https://climatecommunication.yale.edu/publications/sassy-guidebook-2025/>

(Review). Read the description and example for **#audience** in the HC Handbook

<https://my.minerva.edu/academics/hc-resources/hc-handbook/audience/>

Hullinger, J. (2025, June 17). *Fun ways to ditch Fast Fashion for a Sustainable Wardrobe*. Scientific American.

<https://www.scientificamerican.com/article/the-fast-fashion-backlash-is-fueling-a-sustainability-revolution/>

Session 22: (13.1) Data-driven Science Policy and Communication

Learning Outcomes

#interventionalstudy: Design and interpret experimental studies. (C) EA

#observationalstudy: Design and interpret observational studies. (C) EA

#dataviz: Interpret, analyze, and create data visualizations. (C) EA

#gapanalysis: Identify and evaluate whether there are suitable existing solutions to a problem or whether a creative new solution is required. (C) EA

#plausibility: Evaluate whether hypotheses are based on plausible premises or assumptions. (H) EA

#evidencebased: Identify and appropriately structure the information needed to support an argument effectively. (H) MC

Reading, Videos, and Other Preparation Resources

What is the carbon footprint and how is it calculated?. (n.d.). enel.

<https://www.enel.com/learning-hub/energy-transition/carbon-footprint>

The Global Protocol for Community-Scale Greenhouse Gas Emission Inventories (GPC). (2014, December 1). c40knowledgehub.org.

https://www.c40knowledgehub.org/s/article/The-Global-Protocol-for-Community-Scale-Greenhouse-Gas-Emission-Inventories-GPC?language=en_US

Gibbs, D., Harris, N., & Pool, J.-R. (2022, July 28). *Global Protocol for Community-Scale Greenhouse Gas Inventories: Supplemental Guidance for Forests and Trees*. World Resources Institute.

<https://www.wri.org/research/global-protocol-community-scale-greenhouse-gas-inventories-guidance-forests-trees>

2022 San Francisco Sector-based Greenhouse Gas Emissions Inventory At-A-Glance. (2024, April 1). sfenvironment.org.
<https://www.sfenvironment.org/media/14349>

Session 23: (13.2) Communicating Science to a Specific Audience

Learning Outcomes

- #audience:** Tailor oral and written work by considering the situation and perspective of the people receiving it. (H) MC
- #plausibility:** Evaluate whether hypotheses are based on plausible premises or assumptions. (H) EA
- #evidencebased:** Identify and appropriately structure the information needed to support an argument effectively. (H) MC

Reading, Videos, and Other Preparation Resources

- Communication Fundamentals*. (2022). AAAS.
<https://www.aaas.org/resources/communication-toolkit/communication-fundamentals>
- Barron, M., Hagen, A., Ready, E., & Urban, J. (2024). The Science Communicator's Toolkit. American Society for Microbiology.
<https://asm.org/magazine/2024/fall/science-communication-toolbox>
- Hendricks, R. (2017, November 2). *Communicating Climate Change: Focus on the Framing, Not Just the Facts*. Climate Reality Project.
<https://www.climateRealityProject.org/blog/communicating-climate-change-focus-framing-not-just-facts>
- Partnership for Interdisciplinary Studies of Coastal Oceans. 2007.
The Science of Marine Reserves (2nd Edition, United States Version).
www.piscoweb.org. 22 pages.
https://course-resources.minerva.edu/uploaded_files/mu/00405029-0100/science-of-marine-reserves.pdf

Session 24: (14.1) In-class presentations

Learning Outcomes

- #audience:** Tailor oral and written work by considering the situation and perspective of the people receiving it. (H) MC
- #expression:** Interpret, evaluate, and utilize nonverbal communication. (H) MC
- #evidencebased:** Identify and appropriately structure the information needed to support an argument effectively. (H) MC

Reading, Videos, and Other Preparation Resources

- Climate Action Plan 2021 | San Francisco Environment Department (SFE)*. (n.d.).
<https://www.sfenvironment.org/media/14441>

Session 25: (14.2) In-class Presentations

Learning Outcomes

- #conformity:** Identify and analyze social norms, their features, origins and development; evaluate the appropriateness of conformity and contrarianism and apply strategies of social change when applicable. (H) CX
- #comparisongroups:** Identify and evaluate appropriate comparison groups for empirical study designs. (C) EA
- #interventionalstudy:** Design and interpret experimental studies. (C) EA
- #observationalstudy:** Design and interpret observational studies. (C) EA
- #sampling:** Design effective sampling methods and evaluate the interpretation of results accordingly. (C) EA

#studyreplication: Evaluate and incorporate replicability in empirical study designs. (C) EA

#dataviz: Interpret, analyze, and create data visualizations. (C) EA

#modeling: Recognize how models can be used to describe a system, explain a set of data, and generate predictions. (C) EA

#biasidentification: Identify and explain how cognitive biases result from psychological mechanisms or use of heuristics. (C) EA

#biasmitigation: Identify methods to mitigate the effect of biases and determine when it is appropriate to do so. (H) EA

#plausibility: Evaluate whether hypotheses are based on plausible premises or assumptions. (H) EA

#evidencebased: Identify and appropriately structure the information needed to support an argument effectively. (H) MC

Reading, Videos, and Other Preparation Resources

Assessment

Assessing Learning Outcomes

Course grades are based on outcome scores (HC scores) assigned according to a [general mastery rubric](#):

- ① **No Evidence** - No measurable evidence of the learning outcome is presented. The work is missing or was not attempted.
- ② **Not Assessable** - Minimal understanding of the learning outcome is demonstrated. The work somewhat engages with the prompt requirements, but is largely incomplete, contains a substantial flaw or omission, or has too many issues to justify correcting each one. Below passable work.
- ③ **Review Needed** - Passable but partial understanding of the learning outcome is demonstrated, but there are noticeable gaps, errors, or flaws that limit the application scope and depth. The work needs further review or considerable improvement before meeting expectations.
- ④ **On Target** - Understanding of the learning outcome is evident. Additional effort spent on revisions or expansions could improve the quality of the work, but any gaps, errors, or flaws that remain are not critical to the application.
- ⑤ **Excellent** - Understanding of the learning outcome is evident through clear, well-justified work at an appropriate level of depth. There are no remaining gaps, errors, or flaws relevant to the application. The work is strong enough to be used as an exemplar in the course.
- ⑥ **Profound** - Work uses the learning outcome in a productive and meaningful way that is relevant to the task and goes well beyond the stated and implied scope.

Students will receive HC scores on their **assignments** and **in class**. The weightings for these components are specified in the course breakdown above.

- **Assignments:** Assignments are scored on a set of targeted HCs (i.e., from one unit of a course) and may sometimes allow for non-targeted HC applications. The assignment instructions typically include HC guidelines, explaining which HCs are targeted on the assignment, and sometimes offering tips for strong applications. In addition, students must draw on the course resources and classroom discussions to understand the expectations for applying HCs well in the assignments.
- **In-class:** Students will receive regular written feedback and scores every week, including in-class polls, pre-class work, verbal contributions during activities, and breakouts. Students will typically receive one HC score per class session on one of these components, but there may be multiple scores added to some classes. The type of feedback provided on these classroom components can vary, and can include feedback on your participation, your understanding of the material, the clarity of your explanations, how effectively you collaborate, and guidance for upcoming assignments or future classes. See the section below about Classroom Policies to learn more about expectations for Cornerstone courses.

Grades

Each Cornerstone course is viewed as a year-long course that spans two semesters. The final course score for each semester is based on the average scores earned on each assignment and class according to the percentage breakdown outlined near the top of the syllabus. To determine the score for a given assignment, simply take the average of all HC scores given on that assignment. For example, if three HC scores were given on an assignment, then the score for that assignment would be $(\text{score1} + \text{score2} + \text{score3})/3$. The assignment instructions typically outline which HCs are targeted for each assignment.

First Semester Provisional Grades

At the end of the first semester, students will receive a provisional Pass, In Progress, or No Pass grade on their transcript according to the following scale:

Min score ($\geq$)	Max Score ($<$)	Provisional Grade
2	5	Pass
1.7	2	In Progress
0	1.7	No Pass

- **Pass:** 2 or greater
- **In Progress:** between 1.7 and 2. Students in this range will be permitted to continue in the second semester course, but will be placed on Academic Probation and will need to improve their scores in the following semester.
- **No Pass:** below 1.7. Students who do not pass will be placed on Academic Probation and will not be permitted to take the corresponding Cornerstone course in the following semester. They will need to retake the course in the following year. Students who are not continuing with Minerva and will not retake the course in the future will receive a letter grade consistent with the table below (D or F). Students who fall below 1.7 in three or more of the four Cornerstone courses will be dismissed from Minerva.

Second Semester Letter Grades

The provisional grades will be updated and replaced at the end of the second semester, at which point a single final course score will be calculated by combining the course scores from each semester with equal weight: $(\text{fall_score} + \text{spring_score})/2$. This score will then be converted to a final letter grade according to the following table and reflected on students' transcripts for both semesters.

Course Score (lower bound)	Letter Grade	GPA Equivalent
4.00	A+	4.0
3.55	A	4.0
3.35	A-	3.7
3.15	B+	3.3
3.00	B	3.0
2.70	B-	2.7
2.30	C+	2.3
2.00	C	2.0
1.70	C-	1.7
1.00	D	1.0
0	F	0

However, fall and spring scores will only be averaged if the scores for each semester individually are above 1.7. Students with course scores below 1.7 in any individual semester will earn "No Pass" on their transcript for that semester, be placed on Academic Probation, and will need to retake the course in the following year.

- Students who receive below 1.7 in the spring semester, but above 1.7 in the corresponding fall semester course, will retain the "Pass" or "In Progress" on their transcript for the fall and do not need to retake the fall semester course. This can eventually be updated to a letter grade upon retaking and passing the spring semester course.
- Students who are not continuing with Minerva and will not retake the course in the future will receive a letter grade consistent with the table above (D or F).

Importantly, after averaging the scores for both semesters, students must earn a final score of 2 or greater in all four Cornerstone courses (letter grade of C or higher) to fulfill prerequisite requirements for Core courses in subsequent years. The information below outlines possible scenarios for students who do not pass:

- Students with final scores below 2 in one Cornerstone course will need to retake one or both semesters of the course during their second year. They will need to email academicaffairs@minerva.edu and request to work with an advisor to support their success in retaking the failed Cornerstone course in the next year, alongside Core courses.

- Students with final scores below 2 in two Cornerstone courses may email academicaffairs@minerva.edu and request to work with an advisor to support their success in retaking both courses in the following year, but will not be permitted to take any Core courses alongside the retakes.
- Students who fail to pass three or more of the four Cornerstone courses will be dismissed from Minerva.
- Students who obtain scores below 2 and do not intend to continue with Minerva's undergraduate program will receive a letter grade on their transcript consistent with the table above (D or F).

Cornerstone Policies

Please read the general Minerva course policies listed here: <https://my.minerva.edu/academics/policies/course-policies/>.

Cornerstone classes will follow all of the same policies, including professional behavior, attendance, and assignment submission. Those policies will not be re-stated here. Instead, this section clarifies the specific Cornerstone policies for pre-class work, class participation, and makeup work.

AI Usage Policy

The Cornerstones will follow the general policy for AI usage explained on this myMinerva page: [Learning with AI at Minerva](#). In particular:

- Unless otherwise specified by the professor, students may use AI as a learning partner for brainstorming, outlining, editing, and study aids.
- Students may not use AI to generate full sections or paragraphs of text for assignments, pre-class work, makeup work, or any other submitted work unless explicitly permitted.
- Whenever AI is used, students are required to disclose how exactly it was used and provide a link to the relevant chat or session.
- Fundamentally, students must own their work. This means that students remain fully responsible for the accuracy, professionalism, and integrity of all submissions. They are expected to understand all work they submit, regardless of whether AI was used in its creation.
- Violations of this policy will be treated as violations of the honor code and may be referred to the Academic Standards Committee (ASC).

The points above apply to all components of Minerva classes, including pre-class work, breakouts, polls, makeup work, and assignments. As noted below, Cornerstone assignments will include a dedicated **Acknowledgments** section, which is intended as a place to disclose if and how AI was used to work on the assignment. While there may not be an explicitly designated acknowledgment "section" in all pre-class work or in-class resources, the above guidelines for AI disclosure always apply.

Acknowledgments in Assignments

To uphold transparency, academic integrity, and reflection on your learning processes, all Cornerstone assignment submissions must include an **Acknowledgments** section. This section documents the resources, people, and tools (*including generative AI*) that contributed to the final work. This should be a 100–200-word note near the end of your assignment, before the reference list, that explains who or what supported your work, focusing on how it improved the resulting product or your conceptual understanding. The goal of this requirement is not only to ensure integrity, but also to promote meaningful reflection on how tools and collaboration shape your work. Whether you use AI, consult a peer, or reread class notes, knowing how you learn is as important as what you produce. Further, this also allows professors to give you useful feedback on your learning process. Think of this as a chance to demonstrate #selfawareness, #responsibility, and apply principles from the #scienceoflearning. **Please follow the guidelines for what to include in your Acknowledgements on this page.** Failure to submit a complete and accurate acknowledgment may result in grade penalties, resubmission requirements, or an academic integrity investigation.

Classroom Policies

Class Preparation

The effectiveness of Minerva's active learning model relies on all participants coming prepared, so class preparation is an essential part of studying at Minerva. This is usually your first introduction to the topic, and it will take focus and planning to work through. The goal is for everyone to come to class with enough foundation in the material to engage in productive group discussions, exercises, and problem-solving activities. You are not expected or encouraged to master everything before class. However, we expect an honest effort and productive struggle to start a solid base on which you can build during class.

Class preparation usually involves multiple components, and you'll be guided through each one.

- The **study guide** will help you navigate the class preparation, usually containing prompts or questions to address, practice problems, terminology to learn, and other guidance for your notes.
- There is a list of **readings**, videos, and other resources to engage with, each one coming with a description about why it's useful and what to focus on. Use these to complete the study guide.

- There is a separate section on the class page for **pre-class work**, which usually has a workbook for you to complete. This work must be submitted before class. The exercises in the pre-class workbook have been pulled out from the study guide and require evidence of your engagement. Often, the tasks in the pre-class workbook will be used directly during the class, so they are essential to complete for you to be able to participate effectively in class. In other cases, the pre-class work can contain opportunities to reflect on your learning, make progress on an assignment, or receive feedback on your work. **You may receive rubric scores and comments on your effort in the pre-class workbook**, but it will not be graded for every class.

Even though not all pre-class workbooks will be graded, all classes with pre-class workbooks will be checked for completion by Professors and TAs, and **a penalty score will be applied for grossly inadequate pre-class work**. The work need not be flawless, but students are expected to demonstrate evidence of genuine engagement. This means that if you are stuck on an exercise, you should at least explain how you are thinking about it, what you've tried, and what you are confused about. Remember that professors and TAs are available to assist you during office hours, and students should take advantage of this support. Note that penalty scores for incomplete pre-class work will not be applied for absent students, but will still be applied for students who are late to class.

Please note that most of the time, **doing the tasks in the pre-class workbook alone will not be sufficient to prepare for the lesson**. The **prompts in the study guide are equally important** in developing your understanding of the material and helping you enter the class with confidence.

Class Participation

Please see the [main Minerva course policies page](#) regarding general expectations for professional class behavior, including computer requirements and audio-only considerations.

To get the most out of Minerva's education, students must participate actively throughout each class. Our classes are the heart of our pedagogy, designed for students to deepen their understanding of the topic at hand, while simultaneously building a variety of skills, including teamwork, communication, responsibility, and critical thinking. In the Cornerstones, to help everyone build these skills and improve their class participation, students will receive feedback and scores on their classroom engagement. The video recording will be used for this purpose. The feedback and scores will typically go in the "Verbal Contributions" category of the classroom scores, comprising a certain percentage of the course score according to the breakdown set above. The type of feedback on verbal contributions can vary from class to class, and from student to student. The following are common areas you'll receive feedback, but there may be other aspects:

- Understanding of a targeted HC or learning goal, demonstrated through clear, accurate, thoughtful verbal contributions
- Holistic class participation, including asking questions, taking guesses, raising hands, or engaging productively in the class chat (#ClassParticipation)
- Verbal communication skills, including clarity and concision of explanations (e.g., #expression, #composition)

While professors will guide everyone through the material, promote communal engagement, and call on students to contribute, **each student is responsible for their own participation**. In other words, it is your responsibility to be an engaged member of the discussion, rather than the instructor's responsibility to force reluctant participation. It is expected that you find ways to engage, whether that be raising your hand to signal that you have something to add to the discussion or a question to raise, helping your breakout group through the exercises, or using the class chat to ask or answer questions. There is no one metric to capture this (e.g., a certain number of contributions or a threshold talk time), and it will also require dedicated class preparation to be effective. Students should refer to this [Approaching the 90-Minute Class](#) resource to understand the purpose of different parts of Minerva classes and how to make the most of them. It's worth noting here that it is possible to participate too much. There is no need to track statistics to confirm if you are an outlier, but you can practice self-awareness regarding how much space you are taking up in the classroom.

Breakouts

Most Minerva classes will have breakouts where students work together to solve problems and learn from each other through peer instruction and feedback. It's important that all students come to class prepared, having done the readings and pre-class work, so they can contribute to the breakout exercises. Please refer to the [Approaching the 90-Minute Class](#) infographic to learn more about the purpose of breakouts and how to engage effectively. Note the tips in this resource: (1) incite equal participation, (2) teaching leads to learning, and (3) talk it out.

For a given breakout activity, only one (randomly selected) room is recorded in the class video. Thus, do not expect scores to be directly attached to verbal contributions during breakout discussions. However, in some Cornerstone classes, you may receive feedback and scores on your progress or participation during breakouts or a subsequent part of the class. Also, some classes are designed with solo breakout rooms, where students work individually on a set of tasks and may receive scores and comments on their work.

Makeup Work

Given the importance of our classes to your learning as described above, it's essential for students to catch up with the missed material. Skipping this could leave you lost in future classes and disadvantaged on assignments. **In all Cornerstones courses, every absence triggers a makeup work requirement that must be completed and submitted within 7 days of the missed session**. The makeup work prompt (see below) is designed for your benefit: to engage with the material, reflect on your learning, and extract practical takeaways. The policy details are as follows:

- Adequate makeup work must be submitted to the correct Google Form within 7 days of the absence, meaning $7 \times 24 = 168$ hours from the start time of the missed class.

- Submissions will be monitored by TAs and Professors, and students will receive written feedback on their understanding.
- Students who do not submit sufficient makeup work by the deadline will receive a penalty score of 0 attached retroactively to the corresponding class session. This is in addition to an absence penalty, if applied (the first two absences are not penalized).
- On-time but inadequate submissions that don't conform to the standards outlined below (e.g., missing required components) will receive a score of 1 or 2 (depending on the level of inadequacy) attached retroactively to the corresponding session.
- Resubmissions after the deadline will not be accepted.

Makeup Work Instructions

All Cornerstones will follow the same instructions. A small number of specified classes may require alternative makeup work tasks. Please check the end of the syllabus and other classroom materials for those instructions.

After doing all the assigned class preparation and watching the video recording of the class*, compile the following 3 items into a single PDF file and submit them in the Google Form. This should take you about the same amount of time or effort as if you attended the class.

1. **Pre-class work:** Submit your pre-class work, comprised of the tasks in the pre-class work workbook. If there is no pre-class workbook for the session, include a screenshot of the Forum class page demonstrating this. If you have already completed the workbook in Forum, please copy the work into your PDF file for reference.
2. **Process reflection.** Write a 100-150 word reflection describing how you caught up with the material from this class, including the readings, study guide, pre-class work, and the session itself. In your reflection, include at least one piece of concrete evidence to support your claims. This could include:
 1. A description of your visit to office hours with a TA or professor: date, time, who, topics discussed, what you learned.
 2. Your personal notes for the class (pasted into the doc or screenshots).
 3. A conversation with an AI to learn about the concepts in the class, where AI is acting like a tutor or quizzier (include a link to the conversation or a screenshot)."
3. **Valuable takeaways and next steps:** In 150-200 words, address the following:
 1. Identify the main **learning goals** for the class (often found on the introductory slide).
 2. For at least one of the learning goals, highlight a **moment or comment** from the video recording that helped you achieve that learning goal. *Include a timestamp and be specific.*
 3. Identify where you have used or where you plan to use the insights from today's class in an upcoming **assignment**. Again, *be specific about how you'll use these lessons.*
 4. **OPTIONAL:** Explain which topic or concept you are **still confused about** and how you will address this confusion. Keep in mind that you might get **feedback** on this makeup work to help with your confusion, so it will pay off to reflect thoughtfully about this. If you can't think of anything, then tell us part of the class material that you are most interested to learn more about and why exactly it interests you.

*In rare cases when the class recording is not available, students must contact their instructor immediately. An alternate recording can usually be provided by the Cornerstone Lead instructor.

Makeup Work Google Form Link

[Empirical Analyses Makeup Work](#)